

Problem

When the elements of an RLC circuit are both magnitude-scaled and frequency-scaled, which quality is unaffected?

- (a) resistor
- (b) resonant frequency
- (c) bandwidth
- (d) quality factor

Step-by-step solution

Step 1 of 1

Step 1, we know, magnitude scaling is the process of increasing all impedances in a network by a factor, the frequency response remaining unchanged.

The impedances of individual elements, R , L and C are given by,

$$Z_R = R, \quad Z_L = j\omega L, \quad Z_C = \frac{1}{j\omega C} \quad (1)$$

The magnitude scaling, we multiply the impedance of each circuit element by a factor K_m and let the frequency remain constant,